

Ethics Portfolio assignment 2 - Dating app Breeze

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This case study is about the ethical issues of dating app Breeze (College voor de rechten van de mens, 2023).

First impression

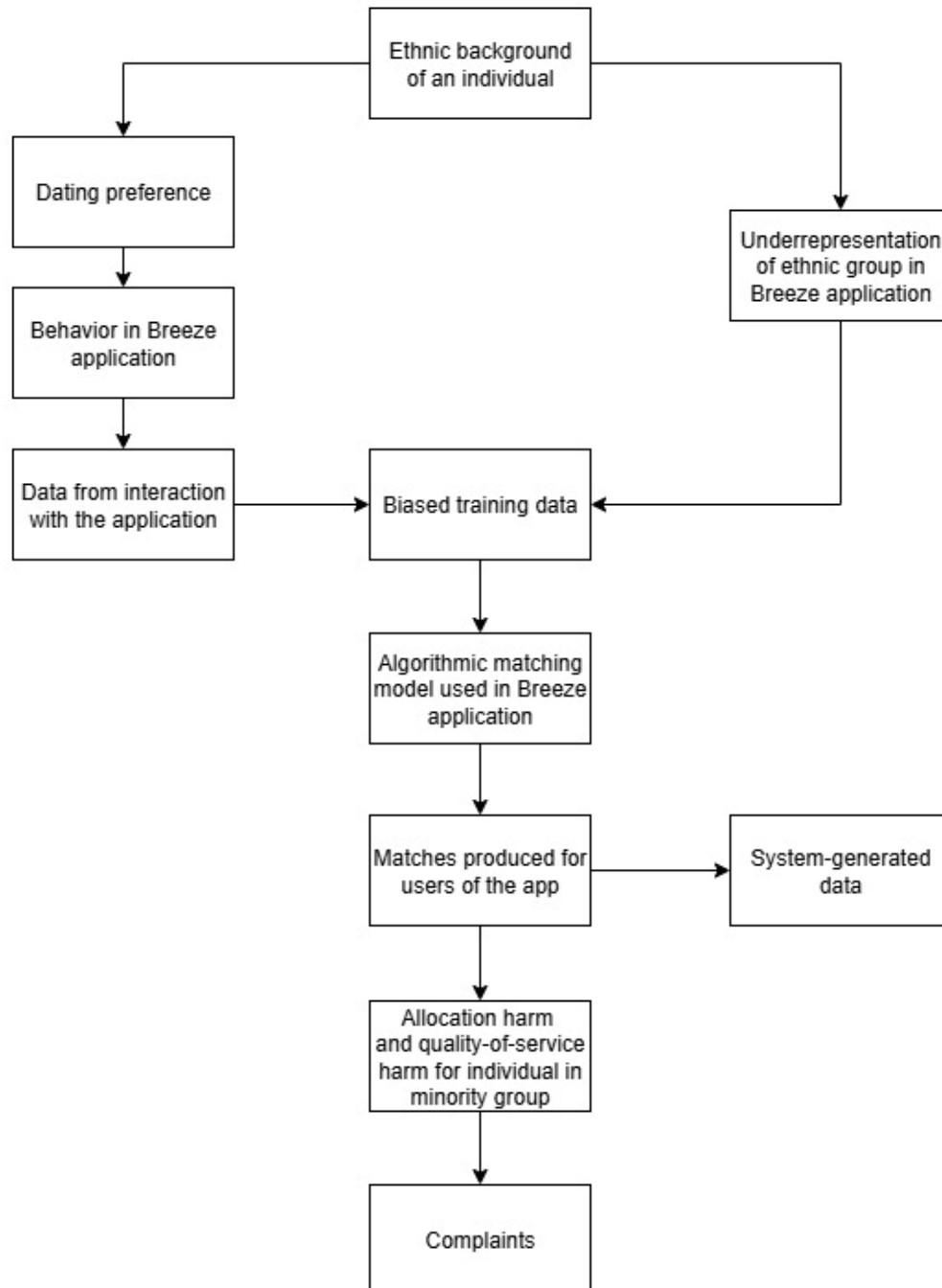
At first sight, Breeze does not appear to intentionally discriminate, but it did identify bias within its matching system. The origin of this bias seems to lie in research that shows that people often prefer partners with a similar ethnic background when it comes to dating and relationships.

Apparently, the system takes existing research on dating preferences into account and also learns from user behavior within the app. As a result, these preferences become part of the training data and can be reinforced by the algorithm over time. This can lead to measurement bias, as historical user interactions reinforce existing patterns. In addition, certain ethnic groups may be underrepresented among the app's users. This underrepresentation can result in selection or representation bias in the data, meaning that the model is trained on data that does not equally reflect all groups.

Based on the study materials from semester 2 of the MADS, this situation can be understood as a case of allocation harm and quality-of-service harm (Weerts, 2025). Allocation harm occurs when individuals receive fewer matches, while quality-of-service harm means that the app works less well for some groups than for others. In this case, these harms became visible through complaints submitted to Breeze, which signaled that certain users experienced unequal treatment.

Directed Acyclic Graph (DAG)

The following DAG was created based on the case of Breeze:



Reflection: Aspects missed in the first impression

Constructing the DAG provided a clearer overview of the situation. It helped to identify several additional insights:

- Multiple separate factors (user behavior and underrepresentation) come together at one important point: The training data. Together, these factors reinforce the bias within the algorithm.
- The system continuously generates new interaction data through its matches. This data can and should be used to monitor how the algorithm performs for different groups.
- Complaints occur after harm has already taken place. They should therefore be treated as important signals that allow the system to be improved across all user groups.

Recommendations for similar cases

For similar cases, I would recommend the following steps to improve the development of the application:

- Use group fairness metrics to compare outcomes between ethnic groups. Possible metrics are demographic parity, equalized odds or equal calibration (Weerts, 2025). These metrics are used to measure the extent to which a particular metric differs across groups of individuals.
- Besides the group fairness metrics, it is also recommended to implement continuous monitoring of model performance.
- Regularly audit training data for underrepresentation or skew.
- Integrate complaints and user feedback into the development process of the application and/or algorithm.
- Test the system under multiple configurations and evaluate performance separately for different demographic groups.

Reference

College voor de rechten van de mens. (2023, september). *Dating-app Breeze mag (en moet) algoritme aanpassen om discriminatie te voorkomen*. Opgehaald van Mensenrechten.nl:
<https://www.mensenrechten.nl/actueel/nieuws/2023/09/06/dating-app-breeze-mag-en-moet-algoritme-aanpassen-om-discriminatie-te-voorkomen>

Weerts, H. (2025, february 17). *Algorithmic Fairness*. Opgehaald van Responsible Machine Learning:
<https://hildeweerts.github.io/responsiblemachinelearning/fairness/introduction.html>